GROUP FUNCTOR REPRESENTABILITY FROM AN OPEN TRANSVERSAL SUBFUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1104-1121 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1123-1132, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_pic.json (pb=566, stmt 1104-1121, proof 1123-1132).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-criterion' also exists in stacks-flat — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01JI: sources/stacks-pic/pinned-original-source.tex, source lines 1132–1132. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JJ: sources/stacks-pic/pinned-original-source.tex, source lines 1132–1132. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let $G : (Sch/k)^{opp} \to Groups$ be a functor. With terminology as in Schemes, Definition 01JI, assume that*

(1) *G satisfies the sheaf property for the Zariski topology,*
(2) *there exists a subfunctor $F \subset G$ such that*
   (a) *F is representable,*
   (b) *$F \subset G$ is representable by open immersion,*

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(c) *for every field extension K of k and $g \in G(K)$ there exists a $g' \in G(k)$ such that $g'g \in F(K)$.*

Then G is representable by a group scheme over k.

Proof. This follows from Schemes, Lemma 01JJ. Namely, take $I = G(k)$ and for $i = g' \in I$ take $F_i \subset G$ the subfunctor which associates to T over k the set of elements $g \in G(T)$ with $g'g \in F(T)$. Then $F_i \cong F$ by multiplication by g'. The map $F_i \to G$ is isomorphic to the map $F \to G$ by multiplication by g', hence is representable by open immersions. Finally, the collection $(F_i)_{i \in I}$ covers G by assumption (2)(c). Thus the lemma mentioned above applies and the proof is complete. □